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(54) **APPARATUS FOR MEASURING
INSULATION RESISTANCE**

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(52) **U.S. Cl.**

CPC **G01R 31/12** (2013.01); **G01R 27/08** (2013.01); **G01R 31/3842** (2019.01)

(58) **Field of Classification Search**

CPC G01R 31/12; G01R 31/3842; G01R 31/52; G01R 27/08; G01R 27/025; G01R 1/203; G01R 1/206; B60L 3/0046

See application file for complete search history.

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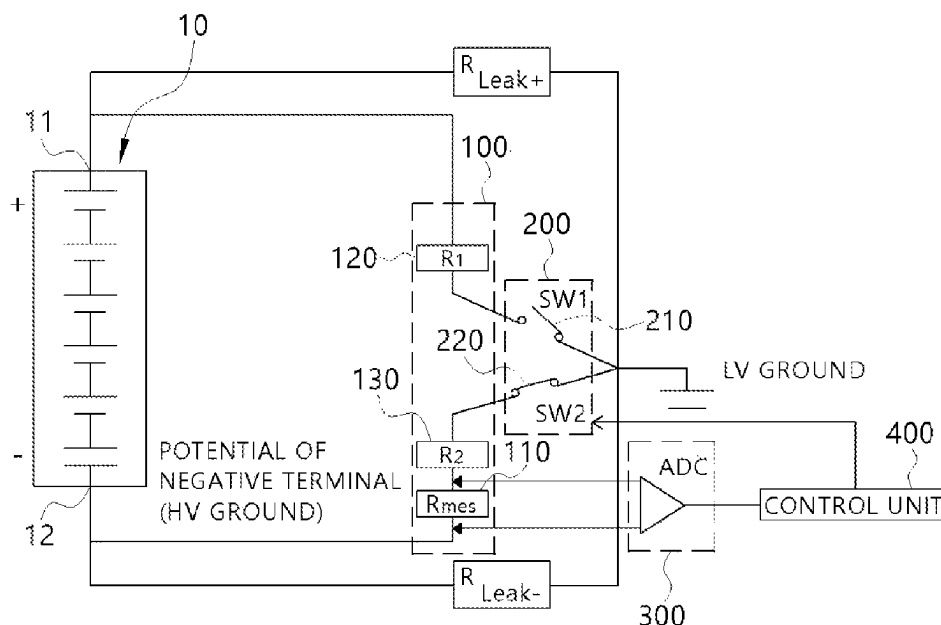
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(57)

ABSTRACT

The present invention relates to an apparatus capable of measuring an insulation resistance of a battery, and according to the apparatus capable of measuring the insulation resistance of the present invention. A voltage measurement unit may measure a voltage with respect to a potential of a negative terminal of a battery rather than the ground, and thus parameters requiring a high precision may be measured using one device, thereby reducing cost, and, because a first sensing substrate and a second sensing substrate are separated, different potentials may be used as the ground, thereby securing a better insulation performance between the first sensing substrate that is an HV side and the second sensing substrate that is an LV side.

14 Claims, 6 Drawing Sheets



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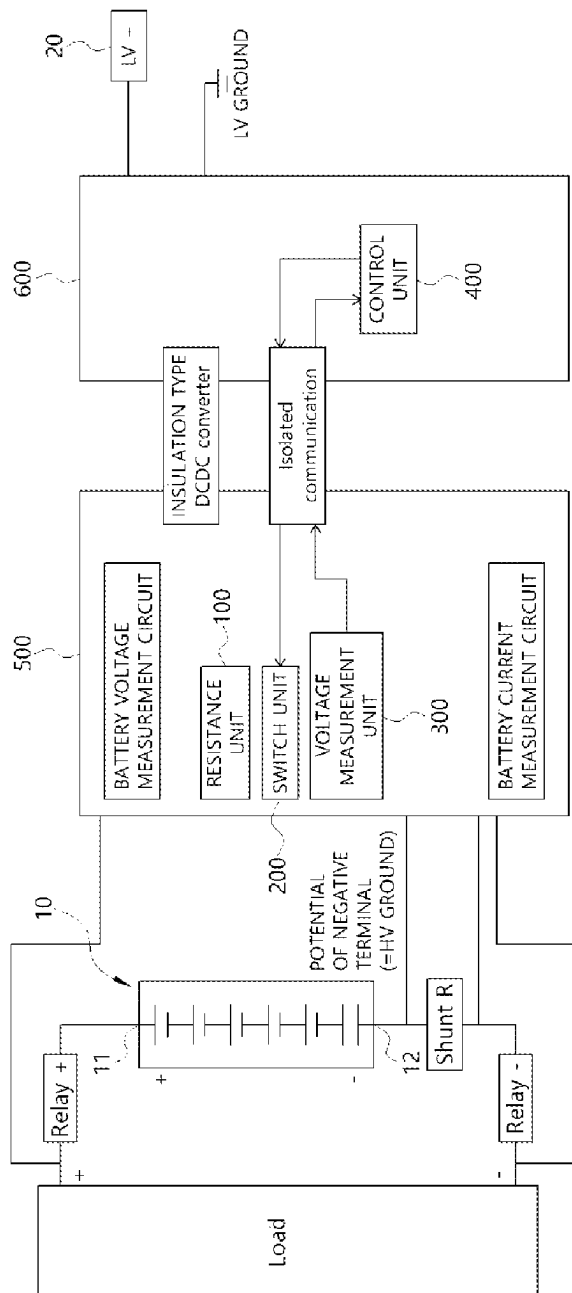


FIG. 2

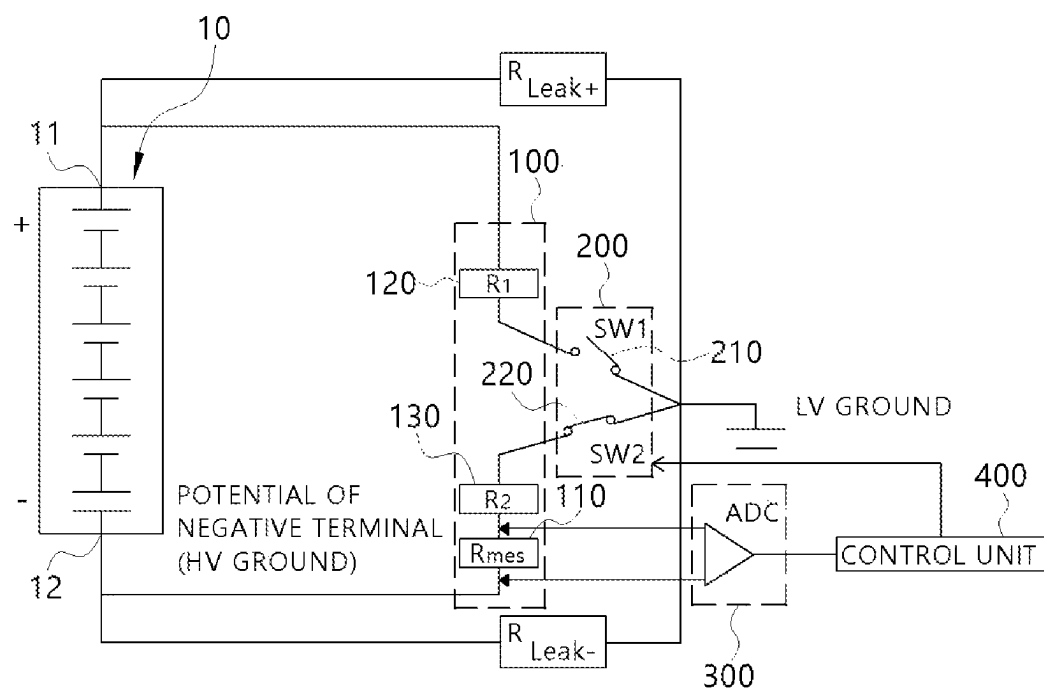


FIG. 3

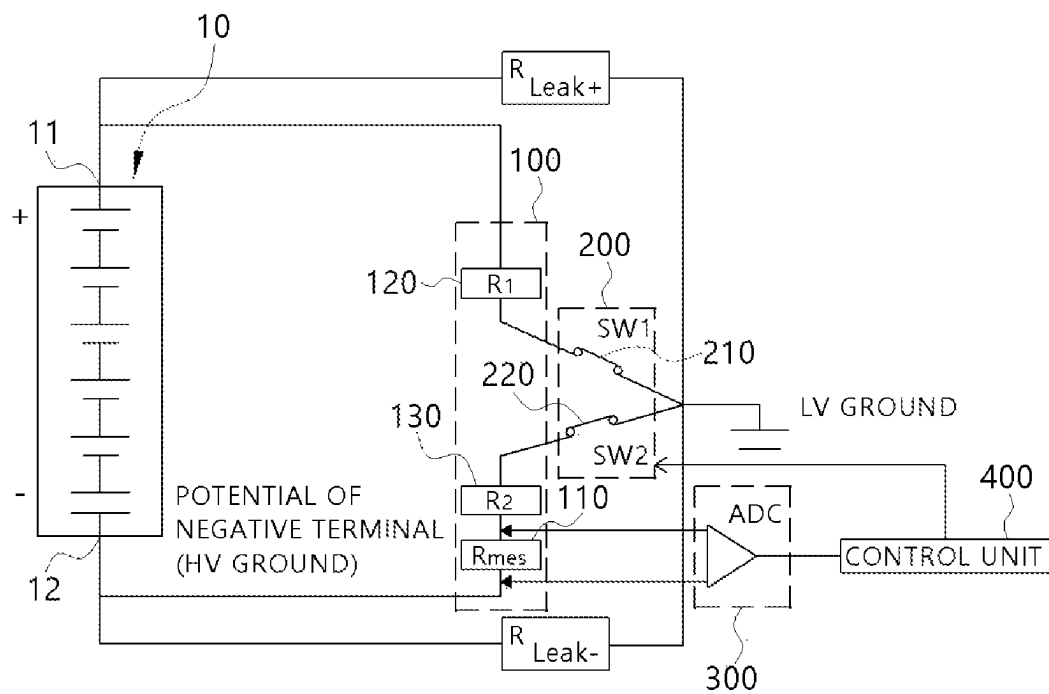


FIG. 4

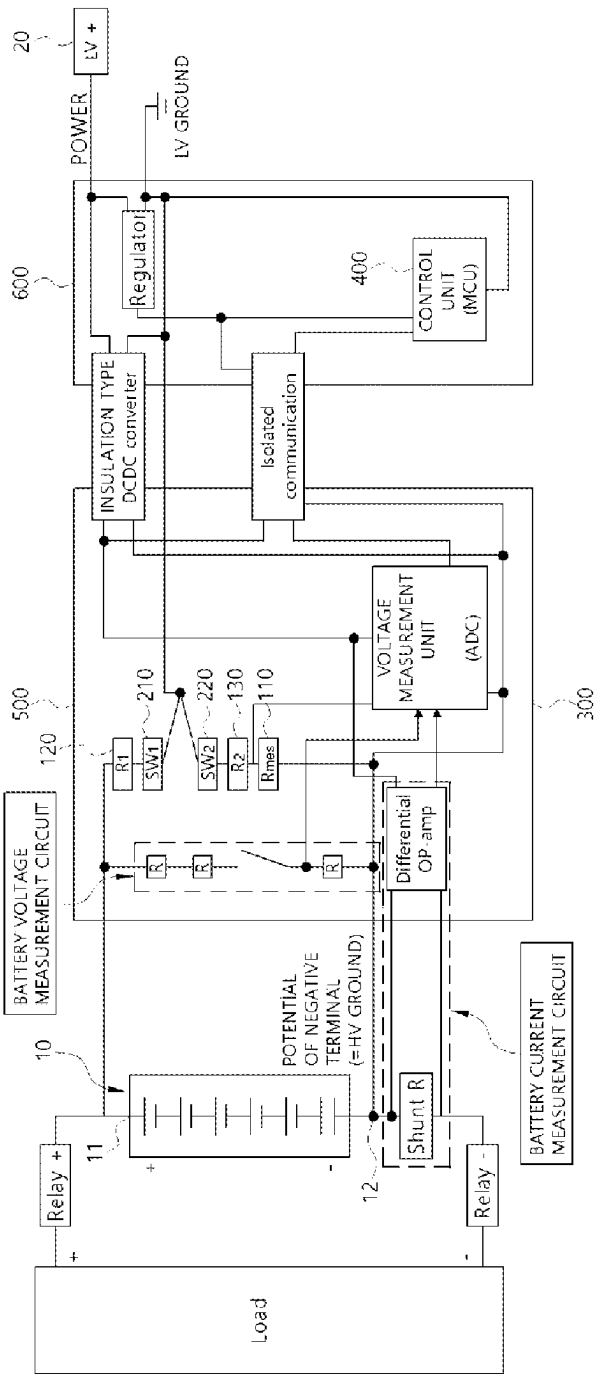


FIG. 5

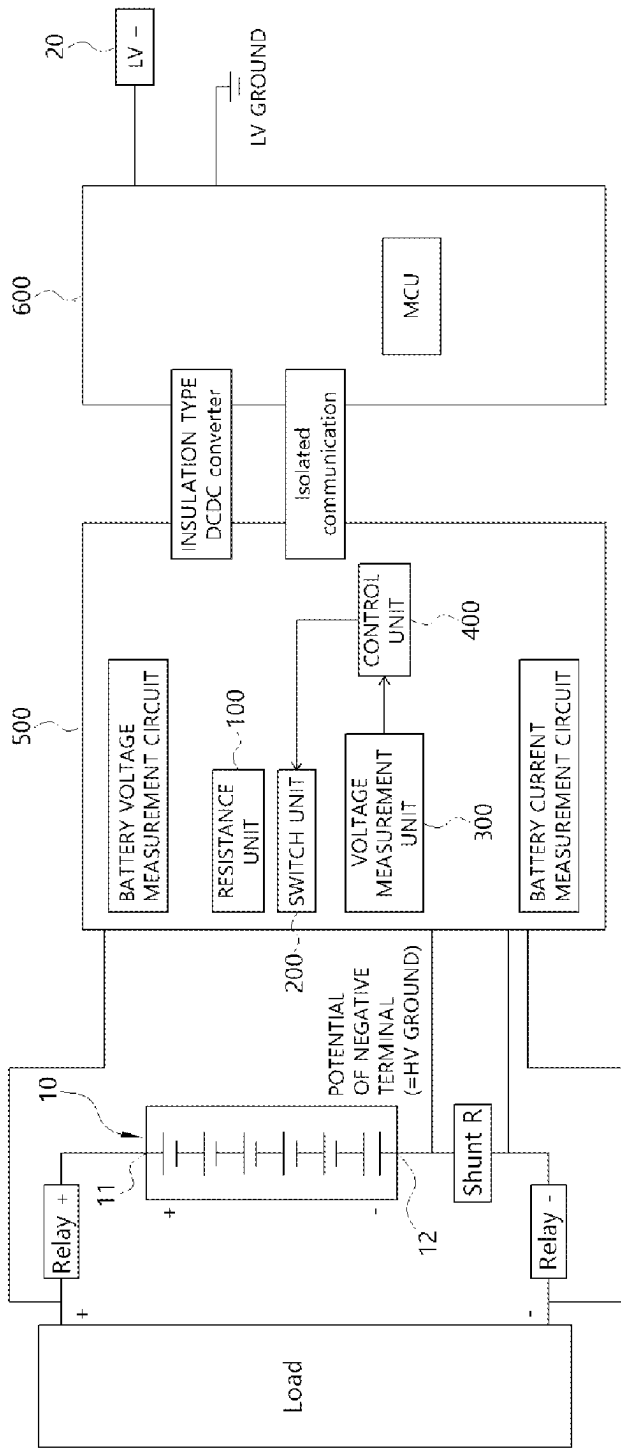
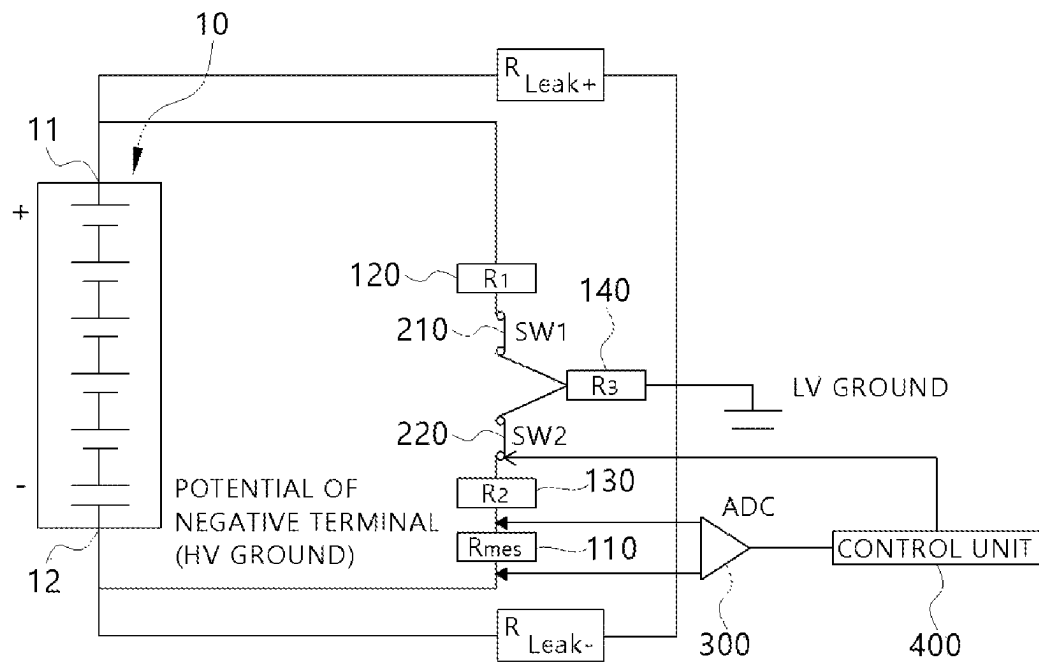


FIG. 6



1

APPARATUS FOR MEASURING
INSULATION RESISTANCECROSS-REFERENCE TO RELATED
APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 17,226,470, filed Apr. 9, 2021, which claims priority to Korean Patent Application No. 10-2020-0044425, filed Apr. 13, 2020, the disclosures of which are hereby incorporated by reference in their entirety.

BACKGROUND OF THE INVENTION

Field of the Invention

The present invention relates to an apparatus capable of measuring an insulation resistance of a battery.

Description of Related Art

Recently, an electric vehicle (EV) using an electric battery capable of charging and discharging as a power source has been commercialized as an eco-friendly vehicle. The electric vehicle includes a battery pack used as a power source that provides a high voltage (HV) to a driving electric motor, an inverter, a converter, and the like that require the high voltage for the operation of vehicle, and a lead acid battery used as a power source providing a low voltage (LV) to electronics of vehicle.

At this time, the insulation resistance of the battery pack may decrease or the insulation thereof may be destroyed due to a deterioration. In this case, it is very dangerous because a person touching a chassis may be electrocuted by the high voltage of the battery pack, and thus it needs to continuously measure and monitor the insulation resistance of the battery pack so that the insulation resistance is not reduced less than a certain level.

Meanwhile, an insulation resistance measurement circuit of the related art has been measured with respect to a low voltage (LV) side ground (for example, a negative terminal of a lead acid battery, a chassis).

RELATED ART DOCUMENT

Patent Document

KR 2010-1610921B1

SUMMARY OF THE INVENTION

Technical Problem

An object of the present invention is to provide an apparatus for measuring an insulation resistance capable of reducing cost by minimizing the number of measuring devices requiring a high precision.

Another object of the present invention is to provide an apparatus for measuring an insulation resistance capable of better securing an insulation performance between a high voltage (HV) side of a sensing substrate and a low voltage (LV) side of the sensing substrate.

Technical Solution

In one general aspect, an apparatus for measuring an insulation resistance of a battery includes: a resistance unit

2

including a voltage measurement resistor configured to divide and measure a voltage of the battery; a switch unit coupled to the resistance unit and including a first switch for forming a different electrical path; and a control unit measuring a voltage applied to the voltage measurement resistor and monitoring the insulation resistance of the battery, and the control unit may measure the voltage applied to the voltage measurement resistor based on a potential of a negative terminal of the battery.

The control unit further comprising a voltage measurement unit configured to measure the voltage applied to the voltage measurement resistor, the voltage measurement resistor may be coupled to the negative terminal of the battery, and the voltage measurement unit may be coupled to the voltage measurement resistance and the negative terminal of the battery.

The apparatus may further include: a first sensing substrate coupled to the battery; and a second sensing substrate coupled to a lead-acid battery having a lower voltage than the battery, and isolated from the first sensing substrate, and the resistance unit and the switch unit may be mounted on the first sensing substrate, and the control unit may be mounted on the second sensing substrate.

The apparatus may further include: a first sensing substrate coupled to the battery; and a second sensing substrate coupled to a lead-acid battery having a lower voltage than the battery, and isolated from the first sensing substrate, and the resistance unit, the switch unit, the voltage measurement unit and the control unit may be mounted on the first sensing substrate.

The resistance unit may further include: a first resistor coupled to a positive terminal of the battery, and a second resistor coupled between the first resistance and the voltage measurement resistor, and the first switch may be coupled between the first resistor and a ground, and the second switch may be coupled between the second resistor and the ground.

The control unit may receive the voltage applied to the voltage measurement resistor while operating the first switch in an off state and the second switch in an on state, and receive the the voltage applied to the voltage measurement resistor while operating both the first switch and the second switch in the on state.

The control unit may calculate a positive insulation resistance and a negative insulation resistance by combining Equation 1 and Equation 2 below.

$$V_{R_{mes_SW2}} = \text{Pack } V \times \frac{R_{Leak-} \parallel (R_{mes} + R_2)}{R_{Leak+} + R_{Leak-} \parallel (R_{mes} + R_2)} \times \frac{R_{mes}}{(R_{mes} + R_2)} \quad [\text{Equation 1}]$$

$$V_{R_{mes_SW1_SW2}} = \text{Pack } V \times \frac{R_{Leak-} \parallel (R_{mes} + R_2)}{(R_1 \parallel R_{Leak+}) + R_{Leak-} \parallel (R_{mes} + R_2)} \times \frac{R_{mes}}{(R_{mes} + R_2)} \quad [\text{Equation 2}]$$

(where $V_{R_{mes_SW2}}$ denotes a voltage applied to the voltage measurement resistor when the first switch operates in the off state and the second switch operates in the on state, $V_{R_{mes_SW1_SW2}}$ denotes a voltage applied to the voltage measurement resistor when both the first switch and the second switch operate in the on state, Pack V denotes a voltage of the battery, R_{mes} denotes a value of the voltage measurement resistor, R_1 denotes a value of the first resistor, R_2 denotes a value of the second resistor, R_{Leak+} denotes a

3

value of the positive insulation resistance, and R_{Leak-} denotes a value of the negative insulation resistance)

The resistance unit may further include a third resistor coupled between the switch and the ground.

The control unit may receive the voltage applied to the voltage measurement resistor while operating the first switch in an off state and the second switch in an on state, and receive the voltage applied to the voltage measurement resistor while operating both the first switch and the second switch in the on state.

The control unit may calculate a positive insulation resistance and a negative insulation resistance by combining Equation 3 and Equation 4 below.

$$V_{Rmes_SW2} = \text{Pack } V \times \frac{R_{Leak-} \parallel (R_{mes} + R_2 + R_3)}{R_{Leak+} + R_{Leak-} \parallel (R_{mes} + R_2 + R_3)} \times \frac{R_{mes}}{(R_{mes} + R_2 + R_3)} \quad [\text{Equation 3}]$$

$$V_{Rmes_SW1_SW2} = \text{Pack } V \times \left[\frac{1}{R_c} - \frac{(R_a \parallel (R_{Leak-}))}{(R_b \parallel (R_{Leak+})) + (R_a \parallel (R_{Leak-}))} \right] \times \frac{1}{R_a} \times R_{mes} \quad [\text{Equation 4}]$$

$$\left(\text{where Equation 4 satisfies} \right)$$

$$R_a = \frac{(R_1 \times R_3) + (R_1 \times (R_2 + R_{mes})) + (R_3 \times (R_2 + R_{mes}))}{R_1}$$

$$R_b = \frac{(R_1 \times R_3) + (R_1 \times (R_2 + R_{mes})) + (R_3 \times (R_2 + R_{mes}))}{(R_2 + R_{mes})}$$

$$R_c = \frac{(R_1 \times R_3) + (R_1 \times (R_2 + R_{mes})) + (R_3 \times (R_2 + R_{mes}))}{R_3}$$

V_{Rmes_SW2} denotes a voltage applied to the voltage measurement resistor when the first switch operates in the off state and the second switch operates in the on state, $V_{Rmes_SW1_SW2}$ denotes a voltage applied to the voltage measurement resistor when both the first switch and the second switch operate in the on state, Pack V denotes a voltage of the battery, R_{mes} denotes a value of the voltage measurement resistor, R_1 denotes a value of the first resistor, R_2 denotes a value of the second resistor, R_3 is a value of a third resistor, R_{Leak+} denotes a value of the positive insulation resistance, and R_{Leak-} denotes a value of the negative insulation resistance.)

Advantageous Effects

According to an apparatus for measuring an insulation resistance of the present invention, a voltage measurement unit may measure a voltage with respect to a potential of a negative terminal of a battery rather than the ground, and thus parameters requiring a high precision may be measured using one device, thereby reducing cost.

In addition, according to the apparatus for measuring the insulation resistance of the present invention, because a first sensing substrate and a second sensing substrate are separated, different potentials may be used as the ground, thereby securing a better insulation performance between the first sensing substrate that is a HV side and the second sensing substrate that is an LV side.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a block diagram schematically showing an apparatus for measuring an insulation resistance of the present invention.

4

FIG. 2 is a circuit diagram of a first circuit formed in the apparatus for measuring the insulation resistance of the present invention.

FIG. 3 is a circuit diagram of a second circuit formed in the apparatus for measuring the insulation resistance of the present invention.

FIG. 4 is a view showing in more detail a connection structure of the apparatus for measuring the insulation resistance of the present invention.

FIG. 5 is a block diagram showing a case in which a position of a control unit is changed as another embodiment of the apparatus for measuring the insulation resistance of the present invention.

FIG. 6 is a circuit diagram showing a case in which a resistance device is added as another embodiment of the apparatus for measuring the insulation resistance of the present invention.

DESCRIPTION OF THE INVENTION

Hereinafter, the present invention will be described in detail with reference to the accompanying drawings. Detailed descriptions of known functions and configurations that may unnecessarily obscure the subject matter of the present invention will be omitted. In addition, the terms such as '... unit' and 'module' mean a unit that processes at least one function or operation, which may be implemented by hardware or software, or a combination of hardware and software.

FIG. 1 is a block diagram schematically showing an apparatus for measuring an insulation resistance of the present invention, FIGS. 2 and 3 are circuit diagrams of a first circuit and a second circuit formed in the apparatus for measuring the insulation resistance of the present invention, respectively, and FIG. 4 is a view showing in more detail a connection structure of the apparatus for measuring the insulation resistance of the present invention.

Referring to FIGS. 1 to 3, the apparatus for measuring the insulation resistance of the present invention includes a resistance unit 100, a switch unit 200, a voltage measurement unit 300, and a control unit 400. Here, the control unit 400 may include the voltage measurement unit 300.

The resistance unit 100 may include a plurality of resistors to distribute a voltage of a battery 10. At this time, the plurality of resistors include a voltage measurement resistor 110, a first resistor 120 and a second resistor 130, the first resistor 120 is connected(coupled) to a positive terminal 11 of the battery 10, and the second resistor 130 is connected between the first resistor 120 and the voltage measurement resistor 110. Here, the battery 10 may be formed by connecting a plurality of battery cells in series or in parallel.

The switch unit 200 is connected to the resistance unit 100 and includes a first switch 210 and a second switch 220, and may form circuits having different paths according to on-off operations of the first switch 210 and the second switch 220. That is, the switch unit 200 coupled to the resistance unit 100 and including the first switch 210 and the second switch 220 for forming a different electrical path.

Next, the voltage measurement unit 300 is a configuration that measures an voltage applied to the voltage measurement resistor 110, and may include an analog-to-digital converter (ADC) that receives an analog voltage signal and converts the analog voltage signal into digital data. In this case, an operation power of the voltage measurement unit 300 may be supplied from a pack of the battery 10 or may be supplied from a lead acid battery 20 through an insulation type DC-DC converter.

5

Meanwhile, in the apparatus for measuring the insulation resistance of the present invention, the voltage measurement unit **300** measures a voltage applied to the voltage measurement resistor **110** with respect to a potential of a negative terminal **12** of the battery **10**.

Next, the control unit **400** measuring a voltage applied to the voltage measurement resistor **110** and monitoring the insulation resistance of the battery. The control unit **400** is a configuration that controls the switch unit **200** and simultaneously receives the voltage applied to the voltage measurement resistor **110** measured by the voltage measurement unit **300** to calculate a positive insulation resistance and a negative insulation resistance.

More specifically, the control unit **400** may receive the voltage applied to the voltage measurement resistor **110** measured by the voltage measurement unit **300** while the first switch **210** operates in an off state and the second switch **220** operates in an on state, and may receive the voltage applied to the voltage measurement resistor **110** measured by the voltage measurement unit **300** while both the first switch **210** and the second switch **220** operate in an on state. Here, the control unit **400** may be a microcontrol unit (MCU).

In this regard, the positive insulation resistance means a resistance between the positive terminal **11** of the battery **10** and the ground, and the negative insulation resistance means a resistance between the negative terminal **12** of the battery **10** and the ground. The ground is a low voltage (LV) ground, and is generally a chassis of a vehicle.

Hereinafter, a specific connection structure of the apparatus for measuring the insulation resistance of the present invention will be described.

Referring to FIGS. **2** and **3**, one end of the first resistor **120** is connected to the positive terminal **11** of the battery **10** and the other end is connected to one end of the first switch **210**. One end of the second resistor **130** is connected to the voltage measurement resistor **110** and the other end is connected to one end of the second switch **220**. The other ends of the first switch **210** and the second switch **220** are connected to the ground.

At this time, the voltage measurement resistor **110** has one end connected to the negative terminal **12** of the battery **10** and the other end connected to the second resistor **130**. In addition, the voltage measurement unit **300** is connected to each of the negative terminal **12** of the battery **10** and the voltage measurement resistor **110**, thereby measuring the voltage applied to the voltage measurement resistor **110** with respect to the potential of the negative terminal **12** of the battery **10**.

Meanwhile, a voltage and current of a battery, and an inverter voltage (hereinafter referred to as high voltage (HV) measurement parameters), etc. are measured with respect to a potential of a negative terminal of the battery. The HV measurement parameters are parameters for diagnosing the battery and are related to a safe operation of the vehicle, and thus need to be measured with a high precision, which is the same as a measurement of the insulation resistance.

However, as described above, the insulation resistance measurement circuit of the related art is measured with respect to the ground. To explain this more specifically, one end of a voltage measurement resistor is connected to the ground, and accordingly, a voltage measurement unit is configured to measure a voltage applied to the voltage measurement resistor with respect to a ground potential.

Therefore, in the related art, because HV measurement parameters and insulation resistance are measured using different potentials as a reference point, a device for mea-

6

suring the insulation resistance needs to use a different device independent from the measurement device of the above-described HV measurement parameters.

In the present invention, the voltage measurement unit **300** is connected to the negative terminal **12** of the battery **10**, other than the ground, so that the potential of the negative terminal **12** of the battery **10** is used as a reference potential when measuring the voltage applied to the voltage measurement resistor **110**. Accordingly, the voltage measurement unit **300** may be configured by adding a voltage measurement function to a device (e.g., an ADC) used when measuring HV measurement parameters. That is, by implementing the measurement of parameters related to a safe driving of an electric vehicle with one device, the number of devices that require a measurement of a high precision may be reduced. Meanwhile, because the high precision is a factor that increases the cost, a cost reduction effect increases as the number of devices that require the high precision decreases.

As a result, by connecting the voltage measurement unit **300** to the negative terminal **12** of the battery **10** other than the ground, there is an effect of reducing the manufacturing cost in configuring a battery system.

Referring to FIG. **4**, this may be more easily understood. As a reference input of the voltage measurement unit **300** of the apparatus of measuring the insulation resistance of the present invention, the potential of the negative terminal **12** of the battery **10** is connected. Therefore, a voltage measurement circuit of the battery **10**, a current measurement circuit of the battery **10**, and the voltage measurement resistor **110** of the apparatus of measuring the insulation resistance may all be connected to one voltage measurement unit **300**, and thus other measurement unit except for the measurement device included in the voltage measurement unit **300** does not need to have a high precision.

Hereinafter, a method of measuring the insulation resistance using the apparatus of measuring the insulation resistance of the present invention will be described in detail.

First, the control unit **400** may form a first circuit by operating the first switch **210** in the off state and the second switch **220** in the on state as shown in FIG. **2**. Here, operating a switch in the on state means that the switch is electrically conducted, and operating a switch in the off state means that the switch is not electrically conducted but is cut off.

The first circuit includes a closed circuit connecting the battery **10**, the positive insulation resistance, the second resistor **130**, and the voltage measurement resistor **110**, and a closed circuit connecting the second resistor **130**, the voltage measurement resistor **110**, and the negative insulation resistance.

While the first circuit is formed, the voltage measurement unit **300** measures the voltage applied to the voltage measurement resistor **110** and transmits the voltage to the control unit **400**. Meanwhile, when applying a voltage distribution principle to the first circuit, the following Equation 1 is calculated.

$$V_{R_{mes_SW2}} = \text{Pack } V \times \frac{R_{Leak} - \|(R_{mes} + R_2)}{R_{Leak} + R_{Leak} - \|(R_{mes} + R_2)} \times \frac{R_{mes}}{(R_{mes} + R_2)} \quad [\text{Equation 1}]$$

Here, $V_{R_{mes_SW2}}$ denotes a voltage applied to a voltage measurement resistor R_{mes} when the first switch **210** operates in the off state and the second switch **220** operates in the

on state, Pack V denotes a voltage of the battery **10**, R_{mes} denotes a value of the voltage measurement resistor **110**, R_1 denotes a value of the first resistor **120**, R_2 denotes a value of the second resistor **130**, R_{Leak+} denotes a value of the positive insulation resistance, and R_{Leak-} denotes a value of the negative insulation resistance.

Next, as shown in FIG. 3, the control unit **400** may form a second circuit by operating both the first switch **210** and the second switch **220** in the on state.

The second circuit includes a closed circuit connecting the battery **10**, the first resistor **120**, the second resistor **130**, and the voltage measurement resistor **110**, a closed circuit connecting the positive insulation resistance and the first resistor **120**, and a closed circuit connecting the second resistor **130**, the voltage measurement resistor **110**, and the negative insulation resistance.

While the second circuit is formed, the voltage measurement unit **300** measures the voltage applied to the voltage measurement resistor **110** and transmits the voltage to the control unit **400**. Meanwhile, when applying the voltage distribution principle to the second circuit, the following Equation 2 is calculated.

$$V_{R_{mes_SW1_SW2}} = \text{Pack V} \times \frac{R_{Leak-} \parallel (R_{mes} + R_2)}{(R_1 \parallel R_{Leak+}) + R_{Leak-} \parallel (R_{mes} + R_2)} \times \frac{R_{mes}}{(R_{mes} + R_2)} \quad [\text{Equation 2}]$$

Here, $V_{R_{mes_SW1_SW2}}$ denotes a voltage applied to the voltage measurement resistor R_{mes} when both the first switch **210** and the second switch **220** operate in the on state, Pack V denotes the voltage of the battery **10**, R_{mes} denotes the value of the voltage measurement resistor **110**, R_1 denotes the value of the first resistor **120**, R_2 denotes the value of the second resistor **130**, R_{Leak+} denotes the value of the positive insulation resistance, and R_{Leak-} denotes the value of the negative insulation resistance.

In this regard, in Equations 1 and 2, $\parallel$ denotes a symbol indicating a parallel connection of resistances. In addition, $V_{R_{mes_SW2}}$ and $V_{R_{mes_SW1_SW2}}$ are values measured by the voltage measurement unit **300**, and the value R_{mes} of the voltage measurement resistor **110**, the value R_1 of the first resistor **120** and the value R_2 of the second resistor **130** are known resistance values and may be stored in the control unit **400**. Accordingly, because there are two unknowns R_{Leak+} and R_{Leak-} , and Equations 1 and 2 including the unknowns are calculated, the positive insulation resistance and the negative insulation resistance may be calculated by a simultaneous equation.

That is, the control unit **400** may calculate the value R_{Leak+} of the positive insulation resistance and the value R_{Leak-} of the negative insulation resistance by combining Equation 1 and Equation 2 calculated from the first circuit and the second circuit.

In addition, the control unit **400** may determine whether the battery **10** has an insulation breakdown based on the value R_{Leak+} of the positive insulation resistance and the value R_{Leak-} of the negative insulation resistance calculated as described above. In this regard, the control unit **400** may store a threshold value of the insulation resistance which is a reference for determining whether the battery **10** has the insulation breakdown. In addition, the control unit **400** may generate a warning signal or generate a signal to cut off power to a main relay of the battery **10** as a follow-up measure when it is determined that the battery **10** has the insulation breakdown.

Meanwhile, the apparatus for measuring the insulation resistance of the present invention may further include a first sensing substrate **500** and a second sensing substrate **600**.

Here, the first sensing substrate **500** is a substrate connected to the battery **10** having a high voltage to sense the voltage and current of the battery **10**, and the second sensing substrate **600** is a substrate provided with an MCU and communication components and connected to the lead acid battery **20** having a relatively low voltage compared to the battery **10**. The sensing substrate may be a printed circuit board (PCB).

In the apparatus for measuring the insulation resistance of the present invention, the first sensing substrate **500** may be a substrate separated (isolated) from the second sensing substrate **600**. Here, separating means not only that the first sensing substrate **500** and the second sensing substrate **600** are insulated through an insulating device (for example, a DC-DC converter including a transformer, an isolated (insulated) communication line, etc.) but also that the first sensing substrate **500** and the second sensing substrate **600** are spatially separated different substrates.

At this time, as shown in FIG. 1, the resistance unit **100**, the switch **200**, and the voltage measurement unit **300** may be mounted on the first sensing substrate **500**, and the control unit **400** may be mounted on the second sensing substrate **600**. In this case, the control unit **400** may send an on/off control signal to the switch **200** through an isolated communication line, and receive the voltage applied to the voltage measurement resistor **110** measured by the voltage measurement unit **300** through the isolated communication line.

Alternatively, in the apparatus for measuring the insulation resistance of the present invention, all of the resistance unit **100**, the switch **200**, the voltage measurement unit **300**, and the control unit **400** may be mounted on the first sensing substrate **500** as shown in FIG. 5. In this case, the control unit **400** may be an MCU additionally provided on the first sensing substrate **500**. As described above, when the control unit **400** is provided on the first sensing substrate **500** with other configurations, the control unit **400** may control other measurement circuits in addition to the apparatus for measuring the insulation resistance. For example, in addition to the control of the switch **200**, the control unit **400** may control a battery voltage measurement circuit switch signal or a battery current measurement circuit switch signal. Therefore, the second sensing substrate **600** which is an LV side does not need to insulate the above-described signals and transmit the above-described signals to the first sensing substrate **500** which is the HV side, and components are configured to exchange only a communication signal between the HV side and the LV side, and thus the cost may be reduced.

Referring again to FIG. 4, a voltage measurement circuit of the battery **10**, a current measurement circuit of the battery **10**, and the voltage measurement unit **300** for measuring the insulation resistance are mounted on the first sensing substrate **500**, and the voltage measurement unit **300** measures the voltage using the negative terminal **12** of the battery **10** as a reference potential, and thus the negative terminal **12** of the battery **10** may be used as the ground (HV ground) of the first sensing substrate **500**.

In addition, the second sensing substrate **600** on which a device (an MCU, a communication component, etc.) using a negative terminal of the lead acid battery **20** as the ground are mounted may use a negative terminal of the lead acid battery **20** as the ground (LV ground) of the second sensing substrate **600**.

That is, because the first sensing substrate **500** and the second sensing substrate **600** are separated into different substrates, different potentials (the HV ground and the LV ground) may be used as grounds, and because the voltage measurement unit **300** is mounted on the first sensing substrate **500**, the voltage of the battery **10** having the high voltage is not applied to the second sensing substrate **600**, thereby securing a better insulation performance between the sensing substrate **500** which is the HV side and the second sensing substrate **600** which is the LV side.

In addition, as described above, a device requiring a relatively high measurement accuracy is mounted on the first sensing substrate **500**, and the first sensing substrate **500** is spatially separated from the second sensing substrate **600**, and thus means (for example, a cover for impact protection) protecting the first sensing substrate **500** may be manufactured in a more robust material.

FIG. **6** is a circuit diagram showing an embodiment in which a resistance device is added to the apparatus for measuring the insulation resistance of the present invention.

Referring to FIG. **6**, the resistance unit **100** may further include a third resistor **140** connected to the other end of the first switch **210** and the other end of the second switch **220**.

In this regard, the control unit **400** may form a third circuit by operating the first switch **210** in an off state and the second switch **220** in an on state and form a fourth circuit by operating both the first switch **210** and the second switch **220** in the on state.

Here, the third circuit includes a closed circuit to which the battery **10**, a positive insulation resistance, the third resistor **140**, the second resistor **130**, and the voltage measurement resistor **110** are connected and a closed circuit to which the third resistor **140**, the second resistor **130**, the voltage measurement resistor **110**, and a negative electrode insulation resistance are connected.

In addition, the fourth circuit includes a closed circuit to which the battery **10**, the first resistor **120**, the third resistor **140**, the second resistor **130**, and the voltage measurement resistor **110** are connected, a closed circuit to which the first resistor **120**, the positive insulation resistance, and the third resistor **140** are connected, and a closed circuit to which the third resistor **140**, the second resistor **130**, the voltage measurement resistor **110**, and the negative insulation resistance are connected.

While the third circuit is formed, the voltage measurement unit **300** measures the voltage applied to the voltage measurement resistor **110** and transmits the voltage to the control unit **400**.

When a voltage distribution principle is applied to the third circuit, Equation 3 below is calculated.

$$V_{R_{mes_SW2}} = \text{Pack } V \times \frac{R_{Leak-} \parallel (R_{mes} + R_2 + R_3)}{R_{Leak+} + R_{Leak-} \parallel (R_{mes} + R_2 + R_3)} \times \frac{R_{mes}}{(R_{mes} + R_2 + R_3)} \quad \text{[Equation 3]}$$

Thereafter, while the fourth circuit is formed, the voltage measurement unit **300** measures the voltage applied to the voltage measurement resistor **110** and transmits the voltage to the control unit **400**. Equation 4 below is calculated through the fourth circuit.

$$V_{R_{mes_SW1_SW2}} = \text{Pack } V \times \left[\frac{1}{R_c} - \frac{(R_a \parallel (R_{Leak-}))}{(R_b \parallel R_{Leak+}) + (R_a \parallel R_{Leak-})} \times \frac{1}{R_a} \right] \times R_{mes} \quad \text{[Equation 4]}$$

Here, Equation 4 satisfies

$$R_a = \frac{(R_1 \times R_3) + (R_1 \times (R_2 + R_{mes})) + (R_3 \times (R_2 + R_{mes}))}{R_1}$$

$$R_b = \frac{(R_1 \times R_3) + (R_1 \times (R_2 + R_{mes})) + (R_3 \times (R_2 + R_{mes}))}{(R_2 + R_{mes})}$$

$$R_c = \frac{(R_1 \times R_3) + (R_1 \times (R_2 + R_{mes})) + (R_3 \times (R_2 + R_{mes}))}{R_3}$$

$V_{R_{mes_SW2}}$ denotes a voltage applied to the voltage measurement resistor R_{mes} when the first switch **210** operates in the off state and the second switch **220** operates in the on state, $V_{R_{mes_SW1_SW2}}$ denotes a voltage applied to the voltage measurement resistor R_{mes} when both the first switch **210** and the second switch **220** operate in the on state, Pack V denotes the voltage of the battery **10**, R_{mes} denotes the value of the voltage measurement resistor **110**, R_1 denotes the value of the first resistor **120**, R_2 denotes the value of the second resistor **130**, R_3 is a value of the third resistor **140**, R_{Leak+} denotes the value of the positive insulation resistance, and R_{Leak-} denotes the value of the negative insulation resistance.

That is, the insulation resistance is calculated based on the same principle as the embodiment of FIGS. **2** and **3** except that the third resistor **140** is added.

As described above, according to the apparatus for measuring the insulation resistance of the present invention, a voltage measurement unit may measure a voltage with respect to a potential of a negative terminal of a battery rather than the ground, and thus parameters requiring a high precision may be measured using one device, thereby reducing cost.

In addition, because a first sensing substrate and a second sensing substrate are separated, different potentials may be used as the ground, thereby securing a better insulation performance between the first sensing substrate that is an HV side and the second sensing substrate that is an LV side.

Meanwhile, the embodiment of FIGS. **2** and **3** in which Equations 1 and 2 are calculated and the embodiment of FIG. **6** in which Equations 3 and 4 are calculated are exemplary. Although the present invention is described by the above-described embodiments and drawings, the present invention is not limited to the above embodiments, and various modifications and variations are possible from these descriptions by those of ordinary skill in the art to which the present invention pertains. Therefore, the technical idea of the present invention should be understood only by the claims, and all equal or equivalent modifications thereof belong to the scope of the technical idea of the present invention.

DESCRIPTION OF REFERENCE NUMERALS

- 10**: battery
- 11**: battery positive terminal
- 12**: battery negative terminal
- 20**: lead acid battery
- 100**: resistance unit
- 110**: voltage measurement resistor
- 120**: first resistor
- 130**: second resistor

11

140: third resistor
 200: switch unit
 210: first switch
 220: second switch
 300: voltage measurement unit
 400: control unit
 500: first sensing substrate
 600: second sensing substrate

The invention claimed is:

1. An apparatus for measuring an insulation resistance of a battery, the apparatus comprising:
 a first resistor connected between a positive terminal of the battery and a ground;
 a second resistor connected between a negative terminal of the battery and the ground;
 a voltage measurement resistor connected between a negative terminal of the battery and the second resistor;
 a voltage measurement unit measuring an insulation voltage applied to the voltage measurement resistor;
 a switch unit connected between the first resistor, the second resistor, and the ground, including at least one switch, and forming circuits having different paths according to on/off operations of the switch; and
 a control unit receiving the insulation voltage measured by the voltage measurement unit and calculating a positive insulation resistance between the positive terminal of the battery and the ground, and a negative insulation resistance between the negative terminal of the battery and the ground.

2. The apparatus of claim 1, wherein the voltage measurement unit is connected to each of the negative terminal of the battery and the voltage measurement resistor.

3. The apparatus of claim 2, wherein the voltage measurement unit has one end connected between the negative terminal of the battery and the voltage measurement resistor, and the other end connected between the voltage measurement resistor and the second resistor.

4. The apparatus of claim 2, wherein the switch unit includes a first switch connected between the first resistor and the ground, and a second switch connected between the second resistor and the ground.

5. The apparatus of claim 4, further comprising:
 a third resistor connected between the first switch and the second switch and the ground.

6. The apparatus of claim 1, further comprising:
 a first sensing substrate connected to the battery; and
 a second sensing substrate connected to a low voltage source having a lower voltage than the battery,
 wherein the first sensing substrate and the second sensing substrate are spatially separated and insulated from each other.

7. The apparatus of claim 6, wherein the switch unit, the voltage measurement resistor, and the voltage measurement unit are mounted on the first sensing substrate, and the control unit is mounted on the second sensing substrate.

8. The apparatus of claim 6, wherein the switch unit, the voltage measurement resistor, the voltage measurement unit, and the control unit are mounted on the first sensing substrate.

9. The apparatus of claim 1, wherein the battery includes a plurality of battery cells connected in series or parallel.

10. The apparatus of claim 1, wherein the voltage measurement unit measures the insulation voltage applied to the voltage measurement resistor based on a potential of the negative terminal of the battery.

12

11. An insulation resistance measuring method using an apparatus for measuring an insulation resistance of a battery, wherein the apparatus comprises:

a first resistor connected between a positive terminal of the battery and a ground;
 a first switch connected between the first resistor and the ground;
 a second resistor connected between a negative terminal of the battery and the ground;
 a second switch connected between the second resistor and the ground;
 a voltage measurement resistor connected between the negative terminal of the battery and the second resistor;
 a voltage measurement unit measuring an insulation voltage applied to the voltage measurement resistor; and
 a control unit receiving the insulation voltage measured by the voltage measurement unit and calculating a positive insulation resistance between the positive terminal of the battery and the ground, and a negative insulation resistance between the negative terminal of the battery and the ground, wherein the positive insulation resistance and the negative insulation resistance are calculated using the control unit,

wherein the method comprises:

receiving the insulation voltage measured by the voltage measurement unit while operating the first switch in an off state and the second switch in an on state; and
 receiving the insulation voltage measured by the voltage measurement unit while operating both the first switch and the second switch in the on state.

12. The insulation resistance measuring method of claim 11, further comprising:
 calculating the positive insulation resistance and the negative insulation resistance by combining Equation 1 and Equation 2 below:

$$V_{R_{mes_SW2}} = \text{Pack } V \times \frac{R_{Leak-} \parallel (R_{mes} + R_2)}{R_{Leak+} + R_{Leak-} \parallel (R_{mes} + R_2)} \times \frac{R_{mes}}{(R_{mes} + R_2)} \quad [\text{Equation 1}]$$

$$V_{R_{mes_SW1_SW2}} = \text{Pack } V \times \frac{R_{Leak-} \parallel (R_{mes} + R_2)}{(R_1 \parallel R_{Leak+}) + R_{Leak-} \parallel (R_{mes} + R_2)} \times \frac{R_{mes}}{(R_{mes} + R_2)} \quad [\text{Equation 2}]$$

where $V_{R_{mes_SW2}}$ denotes a voltage applied to the voltage measurement resistor when the first switch operates in the off state and the second switch operates in the on state, $V_{R_{mes_SW1_SW2}}$ denotes a voltage applied to the voltage measurement resistor when both the first switch and the second switch operate in the on state, Pack V denotes a voltage of the battery, R_{mes} denotes a value of the voltage measurement resistor, R_1 denotes a value of the first resistance, R_2 denotes a value of the second resistance, R_{Leak+} denotes a value of the positive insulation resistance, and R_{Leak-} denotes a value of the negative insulation resistance.

13. An insulation resistance measuring method using an apparatus for measuring an insulation resistance of a battery, wherein the apparatus comprises:

a first resistor connected between a positive terminal of the battery and a ground;
 a first switch connected between the first resistor and the ground;
 a second resistor connected between a negative terminal of the battery and the ground;

13

a second switch connected between the second resistor and the ground;

a third resistor connected between the first switch and the second switch and the ground;

a voltage measurement resistor connected between the negative terminal of the battery and the second resistor;

a voltage measurement unit measuring an insulation voltage applied to the voltage measurement resistor; and

a control unit receiving the insulation voltage measured by the voltage measurement unit and calculating a positive insulation resistance between the positive terminal of the battery and the ground, and a negative insulation resistance between the negative terminal of the battery and the ground, wherein the positive insulation resistance and the negative insulation resistance are calculated using the control unit,

wherein the method comprises:

receiving the insulation voltage measured by the voltage measurement unit while operating the first switch in an off state and the second switch in an on state; and

receiving the insulation voltage measured by the voltage measurement unit while operating both the first switch and the second switch in the on state.

14. The insulation resistance measuring method of claim **13**, further comprising:

calculating the positive insulation resistance and the negative insulation resistance by combining Equation 3 and Equation 4 below:

14

$$V_{R_{mes_SW2}} = \text{[Equation 3]}$$

$$\text{Pack } V \times \frac{R_{Leak-} \parallel (R_{mes} + R_2 + R_3)}{R_{Leak+} + R_{Leak-} \parallel (R_{mes} + R_2 + R_3)} \times \frac{R_{mes}}{(R_{mes} + R_2 + R_3)}$$

$$V_{R_{mes_SW1_SW2}} = \text{[Equation 4]}$$

$$\text{Pack } V \times \left[\frac{1}{R_c} - \frac{(R_a \parallel (R_{Leak-}))}{(R_b \parallel (R_{Leak+})) + (R_a \parallel (R_{Leak-}))} \times \frac{1}{R_a} \right] \times R_{mes}$$

where Equation 4 satisfies

$$R_a = \frac{(R_1 \times R_3) + (R_1 \times (R_2 + R_{mes})) + (R_3 \times (R_2 + R_{mes}))}{R_1}$$

$$R_b = \frac{(R_1 \times R_3) + (R_1 \times (R_2 + R_{mes})) + (R_3 \times (R_2 + R_{mes}))}{(R_2 + R_{mes})}$$

$$R_c = \frac{(R_1 \times R_3) + (R_1 \times (R_2 + R_{mes})) + (R_3 \times (R_2 + R_{mes}))}{R_3}$$

$V_{R_{mes_SW2}}$ denotes a voltage applied to the voltage measurement resistor when the first switch operates in the off state and the second switch operates in the on state, $V_{R_{mes_SW1_SW2}}$ denotes a voltage applied to the voltage measurement resistor when both the first switch and the second switch operate in the on state, Pack V denotes a voltage of the battery, R_{mes} denotes a value of the voltage measurement resistor, R_1 denotes a value of the first resistance, R_2 denotes a value of the second resistance, R_3 is a value of a third resistance, R_{Leak+} denotes a value of the positive insulation resistance, and R_{Leak-} denotes a value of the negative insulation resistance.

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